Course BUSI UNI3704-001 — Spring 2019
Making History Through Venturing
Friday, 9:00am - 12:00pm

516 Hamilton Hall

Dr. Amol Sarva
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Course Description

The philosophers have only interpreted the world in various ways;
the point, however, is to change it.
Karl Marx, "Theses on Feuerbach"

Technology-driven startups are making history, an undertaking that used to be for the masters of powerful institutions. Now nearly everyone is involved. This course delves into the four major drivers of how to make history happen, drawn from the emerging doctrine of Silicon Valley entrepreneurship: people, product, market and financing.

We aim to look broadly at commercial, social, technical and non-technical new ventures. Our approach will be rooted in the history and theory of revolutions, movements, discoveries, and innovation — Pericles, Seneca, Machiavelli, Vico, Clausewitz, William James, Marie Curie, Teddy Roosevelt, Grace Hopper, Jennifer Doudna but also Silicon Valley and venture capital, Marc Andreessen, Elon Musk and including Martin Luther King Jr. and civil rights, Margaret Sanger and suffrage/women's health, Otpor and anti-totalitarianism, and modern American Tea Party/Resist. We will compare major competing frameworks of ideas, develop a perspective on the new forces shaping our world and their opportunities, and be better prepared to make wise choices on the field of innovation.

Instructor Bios

Dr. Amol Sarva (CC '98) is a startup founder (Virgin Mobile USA, IPO; Peek, acquired; Halo Neuroscience, VC-backed; Knotable, VC-backed; Knotel, VC-backed). He is an investor in 50+ startups, and a mentor at Techstars and other venture firms and accelerators. He has a B.A. in economics and philosophy from Columbia College, and his Ph.D. dissertation at Stanford was "The Concept of Modularity in Cognitive Science." He has taught, with Damon Phillips, "Venturing to Change the World" at Columbia College since 2016, and formerly taught philosophy at Stanford. His writings on entrepreneurship are at Amol.Sarva.co and in Fortune, Forbes, and Huffington Post; he has given talks at DLD, Darwin's Circle, Maker Faire, and Founders Forum.

Shopping Period and Final Registration

If you are on the waitlist, you may still get in. You are also welcome to shop and decide if it's for you. *Prerequisite:* "Venturing to Change the World" or an equivalent course. To make sure this class provides its students with a valuable experience, we will review and approve the final **25** students for this class. Our aim is for an interesting, qualified, and motivated group.

Course Logic

Making history was once for princes and priests, and since the Renaissance, for generals, politicians, scientists and explorers. Today, innovation and technology are the forces changing the world: inventors, engineers, startup founders, radicals and movements. This puts the power in the hands of upstarts, working on new ideas and building huge new organizations: commercially and/or socially-driven.

We live in the age of Apple, Google, Facebook, Amazon and you. Your idea can change the world.

Building on the survey course (*Venturing to Change the World*), we explore four families of advanced topics in creating successful organizations in an age of accelerating change. As innovation is moving from an accidental, artisanal process to a large-scale societal machinery, there are common methods and problems.

We examine competing ideas from Ancient Greece to Silicon Valley, drawing on philosophy, economics, history, sociology, engineering, finance and management topics that animate powerful commercial and social ventures. We put one foot in the canon, and the other deep in the cutting-edge thinking of frontline practitioners -- venture capitalists, startup founders, and real-time critics.

Upstarts are defined by their choice of people, product, market and financing.

Leadership and creating organizations (People): Building the apparatus of change means creating upstart teams, then scaling them into institutions. The job of leader is unique. We'll examine the techniques of leaders from the Pericles to Gandhi to Jobs.

How ideas become the built-world (Product): Until the mid-20th century, little of the world was designed or engineered and produced for global consumption. Today, virtually every object we encounter in society is a designed and mass-produced artifact. We'll consider the new art and science of product design and engineering management, as well as how to deploy it to create global impact.

Scientific discoveries, innovation and evaluating the potential for new ventures (Market): Rapid change creates opportunities, but selecting and developing the right ones is difficult. We'll examine the innovation process itself, and look deeply at two technology revolutions underway for the next decade (synthetic biology, artificial intelligence).

Economics, finance, and fundraising for innovation and ventures (Financing): Ventures need to work economically, and we'll understand the tools of financial analysis and the social science drivers of business models, fundraising, financial markets, and deals.

This course is not a practicum for trial and error learning. It is a seminar on the theoretical and historical knowledge of how leaders seize opportunities to change the world.

Keywords: science, technology, innovation, management, finance, fundraising, operations research, organizational behavior, ethics, social impact, leadership, philosophy.

Teaching Philosophy

Our first objective for this course: we will give you frameworks and tools to evaluate, initiate, and engage transformative ideas and new ventures. Second, we will push you to think critically. Long after you graduate, extemporaneous and deliberative thinking will continue to reward you professionally and personally. Third, we enjoy combining findings from academic research, classical writing, and first-draft theories from the world of practitioners.

Course Requirements

Given how dynamic the world of venturing is, most of the readings for this course are online articles by experts. The items are relatively short, so the load should be reasonable and the cost low. We will not use a textbook, but there are many distinct volumes we'll distribute.

Three requirements for Making History Through Venturing course: [1] Weekly Discussion Questions, [2] Class Participation, and [3] Group Assignments.

1. **Weekly Discussion Questions**

For each session, you are required to complete discussion questions (posted on CourseWorks) related to the readings. Submissions are due by 11:00 PM on the evening before class. Your answers must be based upon your own reading and analysis. If the class is prepared, the course session works. If not, you are depriving your classmates of a useful perspective in the room. The teaching team will read these beforehand to prepare the day. If you complete the discussion questions you will get full marks. Respect the honor code.

2. **Class Participation**

This isn't a lecture course. We pour our energy into designing this course and have prepared interesting and difficult topics for you to tackle with your colleagues. This participatory process is central to how you process these ideas, and we will hold you accountable for it. Part of what you are learning is participation and leadership itself. We will ask you to chime in, present an idea, conduct a debate, or announce a conclusion — accounting for in-person vs. in-writing preferences.

Your contributions will be evaluated on quality but also on quantity. If you are not in the classroom, you hurt the class. You can have one absence with notice in advance; no more. Be on time; we'll penalize lateness.

We will hold ourselves to the standard of keeping the class engaged as long as it is prepared. Devices are encouraged. Refer to your source materials, notes, and research in real time.

3. **Group Assignments**

The written work for this course includes the various documents and research that a new venture requires in its early formation. You will develop these pieces in steps during the semester, then pull it together into something polished at the end.

You'll work in groups of three students for each of the group assignments . Forming and working in teams is a key to entrepreneurship, and this will be your practice on this. You will change groups several times through the semester, and final grades for all team members will not be uniform.

You will not be married to a single idea for the entire course, and you won't be building your own idea. You will make several iterations on developing a publicly available company's idea. The course is a set of exercises to build and refine your skills dispassionately. (So this isn't a launchpad for your own startup. Do that later.)

There are four group assignments through the semester.

Topic and company selection: At the third assignment, teams will select a venture concept from a fixed dataset of companies. The dataset is any company mentioned for raising venture capital (not private equity or public) financing by the Fortune "Term Sheet" newsletter in the folder provided on Courseworks. If you have a social venture idea linked to these topics, but not on the Term Sheet dataset, please use the teaching team for guidance.

Team formation: Before each of the four assignments described below we will allow students time to form a three-person team. You will need to select different team members for each of the assignments, so the groups will change for each assignment. As teams re-form, you can have at most one member from a previous group assignment as a member of the group for your new assignment. The assignments will require you to evaluate each team member's contribution, such that everyone will receive an individual contribution score.

Group Assignment 1 (due Class 5, February 22): Market Opportunity Analysis Memo

> Write a memo recommending a promising market area for new startups. Discuss changing market landscape and technology, profile competition, consider revenue and behavior changes, forecast long-run change potential, evaluate value-creation opportunity. 3 pages plus up to 2 pages of charts. Use the Fortune Term Sheet newsletter as a guide for example companies, as well as the reading and your own research. Avoid wandering too far off the topic areas contemplated by the course directly.
>
> This is group work. You write as a team.
>
> Use a market area/company type defined by the selection criteria above.

Group Assignment 2 (due Class 7, March 1: Product Vision, Customer Needs and Positioning

> Building on a market shift identified in assignment 1, identify a specific product vision and create a product roadmap to serve it. Develop a hypothesis on the customer market and segmentation, its needs, and outline a product prototype that could meet

those needs. Lay out a roadmap that identifies the first internal prototype, the first beta, the first released product, and how the product family can develop over time. Sketch a prototype or mockup of the product. Identify key questions you seek to answer as you build, test, and iterate.

This can be a 10 page presentation deck, including charts and illustrations.

New group this time.

Same market/space selection criteria. You likely will not be profiling or building on a real company in this example. You may build on your previous group work by picking from your team's previous, various learnings.

Group Assignment 3 (due Class 10, April 5): Product Vision and Strategy, 2.

For this assignment pick a company from the Fortune Term Sheet dataset that has raised at least $10 million in investment.

Once again, we will be making the case and analysis around: customer profile, needs and segmentation; product vision, roadmap; sketches and testing approach for iterating the product. This time you will also lay out a technology plan for your product, with a sequencing of what we build and what steps we you would use to manage it. Include time and budget estimates.

Get information in creative ways. These are real companies.

Same team rotation and market/company rules above.

Group Assignment 4 (due Class 14, May 3): Company Pitch for Fundraising

This is a polished pitch deck for a company that someone developed in assignment 3. You'll build on the product vision, adding sections on customer psychology and marketing, as well as connecting the product idea to a grand technology vision rooted in a long-term trend such as AI or Synth Bio. You'll record a five minute video talking through your basic pitch. We'll watch these, give notes, and then workshop them in class. You'll be graded on what you do in writing and in presenting. This will be a 15-20 page deck.

Same team and market/company rules about rotation and selection set.

Grading Each Group Assignment

Each assignment will be graded on three elements: [1] the learning and progress that has occurred during the group assignment; [2] the diligence with which you have executed assignment; and [3] the quality of the work. We want assignments that have been well developed and iterated, well researched, and convincingly presented. That said, the quality of the venture is not the determinant of your grade, and it is quite possible to have a strong grade if the venture's underlying idea is not high quality.

Please submit all assignments online through CourseWorks.

GRADING

The following weights will be used to calculate grades for the class:

Weekly Discussion Questions	……..25%
Class Participation	…..…….……30%
Group Assignment 1 (Class **5**)	……..10%
Group Assignment 2 (Class **7**)	……..10%
Group Assignment 3 (Class **10**)	……..10%
Group Assignment 4 (Class **14**)	……..15%

At the midpoint of the course (after week 7) you will receive feedback on your performance so that you will have a sense of your progress and trajectory.

Class Agenda

[Field] Real-time writings from frontline practitioners or observers
[Academic] Classical or academic authors

1. METHOD, HOW TO FIND THE TRUTH ABOUT CREATING NEW, LARGE-IMPACT VENTURES.

DATE: January 25
DUE: Reading
TO DO: Get signed up
WEEKLY QUESTION:
AGENDA: The argument of this course is that there is a way to build new organizations around new ideas, a way that can be described, improved and repeated. Many venture courses attempt to have you learn-by-doing. This course will look at and debate the theories that have emerged, how they are different, and how they might evolve.

First week: we look at several grand theories of how it all works. We'll inspect the classic debate on method from philosophy of science.

READING:

[Field] *PMarchive*, Marc Andreessen, (2007, online). The collection of public writings on venture and startups by legendary VC and founder before he launched his venture fund, and after he had steered multiple companies to IPOs. Selections Luck, Psychology, VCs 1,2,3.

[Field][Lecture 1 transcript or video], YC Startup School 2017, open courseware from the premier for-profit company-building center in Silicon Valley YCombinator.

[Academic][Pages 1-26], *High Tech Startup* 2017 Update circular (not the book), John Nesheim (originally 2000, Free Press), the first data-driven detailed chronicler of patterns in financing and success of Silicon Valley across 1000s of companies and 100s of VCs.

[Academic]*Discourse on Method*, Renee Descartes (Project Gutenberg, 1637). First two chapters, regarding the emerging requirements for doing science.

2. PREDICTING THE FUTURE

DATE: February 1
DUE: Reading
TO DO: Form teams. Assignment in class 5.
WEEKLY QUESTION:
AGENDA: Selecting a market means selecting a force of change. Let's understand this.

READING:

[Academic]*Superforecasting*, Philip Tetlock (Crown, 2015). Penn economist reports his research on how the best predictions of the future get made.

[Academic]*Fourth Industrial Revolution*, Klaus Schwab of World Economic Forum (2016, *Foreign Affairs* and Weforum.org)

[Academic]*Nobel Prize Lecture, Chemistry 1911*, Marie Curie (Nobel Foundation)

3. ARTIFICIAL INTELLIGENCE

DATE: February 8
DUE: Reading
TO DO: You should have a team by now.
WEEKLY QUESTION:
AGENDA: We'll delve into one of the biggest forces changing society.

READING:

[Academic]Simon, Newell (1971). "Human problem solving: The state of the theory in 1970". American Psychologist, 26(2), pp. 145. (pdf) (bib)

[Academic]"Part 1" and "Part 26", *Artificial intelligence: a modern approach.* Russell, Norvig, Canny, Malik, Edwards (Pearson, 2010)

[Academic]*The Second Machine Age*, Erik Brynjolfsson and Andrew McAfee (WW Norton, 2016). MIT economists forecast the social and economic impact of labor automation.

[Academic]*Oral History of Captain Grace Hopper* (pages 10-25), Grace Hopper and Computer History Museum (1980).

[Field] "A visual introduction to machine learning" by R2D3. Easy introduction to how machine learning works.

4. SYNTHETIC BIOLOGY

DATE: February 15
DUE: Reading
TO DO: You should be deep into assignment 1.
WEEKLY QUESTION:
AGENDA: We'll delve into one of the *other* biggest forces changing society.

READING:

[Academic]Regenesis, George Church (Basic Books, 2012). Legendary Harvard biologist surveys the technical and commercial possibilities of the next era in biology.

[Field] "The Genesis Engine", Amy Maxmen (*Wired*, 2015).

[Academic]Review of Crack in Creation by Jennifer Doudna and *Reply*, *Nature 2017*, by Nathaniel Comfort and Jennifer Doudna.

[Academic]"Everything you need to know about CRISPR gene editing's monster year", Antonia Regalado, MIT Technology Review, 2015.

5. PRODUCT: DESIGN THINKING

DATE: February 22
DUE: Assignment 1. Reading.
TO DO: Form teams for the next assignment. Only 2 weeks.
WEEKLY QUESTION:
AGENDA: Once you have a market selected, you have to figure out what to do. Designing a product concept and making it happen. We begin with the users and their needs.

READING:

[Academic][Chapters 1,2,7]The Design of Everyday Things,Donald Norman (Basic Books, 1988). First published as the *Psychology of Everyday Things*, cognitive scientist Don Norman defines the agenda for product design.

[Field][Chapters to select] *Art of Innovation*, Tom Kelley. Head of design firm IDEO and professor at Stanford Design School, associated with creation of design-thinking movement.

[Field] "Innovation Trends Design Thinking" by BBVA Innovation Center

[Field] "How a Facebook Designer Thinks", Julie Zhou at Stanford eCorner. (video)

[Field] [Chapters 2 and 3] *Startup Owner's Manual*, Steve Blank & Bob Dorf.

6. PRODUCT: TAKING DESIGN TO SHIPPING

DATE: March 1
DUE: Reading.
TO DO: Next week, assignment 2 is due.
WEEKLY QUESTION:
AGENDA: Once you have a vision, you have to make it happen. This is how.

READING:

[Field] "What I Learned Building Medium (So Far)" by Ev Williams, founder of Medium and Twitter on the process of inventing huge successful platform Medium.
[Field] "We Don't Sell Saddles Here" by Stewart Butterfield, founder of Slack and Flickr, on the vision for his new product on the day of Slack's launch.
[Field] "Good Product Manager/Bad Product Manager", Ben Horowitz, venture capitalist.
[Field] "How to hire a product manager" by Ken Norton of Google. Classic piece defining an ill-defined role.
[Field] "PM at Microsoft" by Steven Sinofsky, former head of Windows and now VC.
[Field] "A dozen lessons about Product Market Fit", Tren Griffin (2016, *25iq)*
[Field] "WWDC 2017 Some thoughts", Steven Sinofsky, (2017, *Learning by Shipping*)

7. PRODUCT ENGINEERING: MANAGING THE ENGINEERING

DATE: March 8
DUE: Reading. Assignment 2 is due.
TO DO: New teams for the next assignment. Start.
WEEKLY QUESTION:
AGENDA: Shipping is one thing. Another thing is managing processes to keep doing it.

READING:

[Academic][Chapters 1,2,3] How to Design Programs, Matthias Felleisen et al. (MIT Press, 2000) Contents, first, second, third.

[Academic][Ch1 "The Tar Pit", Ch 2 "Mythical Man Month, Ch 5 "Second System Effect"] *The Mythical Man-Month*, Frederick Brooks (Addison Wesley Professional, 1975). Early template for the dynamics and patterns of software development engineering projects.

[Field] [Chapters 1, 2. **maybe 3 also**] *SCRUM: The art of doing twice the work in half the time*, Jeff Sutherland (Crown, 2014). The creator of Agile Development lays out the argument.

[Field] [Read all] "Postmortems at Etsy" by Daniel Schauenberg of Etsy, "Continuous Deployment at Etsy" by Ross Snyder, and "How does Etsy manage development and operations" by Chad Dickerson, CEO of Etsy. (2011, published to Qcon Conference, and Etsy's corporate blog)

8. THE NATURE OF FOUNDERS: YOU AS THE FOUNDER

DATE: March 15
DUE: Reading.
TO DO: Your team should be together. Assignment 3 coming soon.
WEEKLY QUESTION:
AGENDA: Now the topic is people, and the people are the people like you. What does it take to be the founder? What happens to you? How should you conduct yourself?

READING:

[Field][Chapters 1, 12, 19, 30] *Founders At Work: Stories of Startups Early Days*, Jessica Livingston (Apress, 2009)

[Academic][Chapter 1]The Effective Executive, Peter Drucker (HarperCollins Business, 1966).

[Academic][Chapters 1, 3-10, 22, 42], *Letters to Lucilius*, Seneca (c. 65 AD)

[Academic]Are You a Disrupter or a Builder?, Dana Kanze & Sheena Iyengar (Harvard Business Review, 2017).

9. LEADING OTHERS IN THE VENTURE

DATE: March 29
DUE: Reading.
TO DO: Work hard: Assignment 3 is next week.
WEEKLY QUESTION:
AGENDA: Working solo is not like working in a team or leading one. We inspect the varied principles of leadership.

READING:

[Academic][Page 88-100 selection] *Leadership without Easy Answers*, Ronald Heifetz (Harvard UP, 1998). Political leadership scholar.

[Field] [Chapter 3-6] *High Output Management*, Andy Grove (Vintage, 1995). Cofounder of Intel on how to build a machine.

[Field] [Chapter selections TBD] *Principles*, Ray Dalio, 2006-2017. Founder of world's largest hedge fund provides his method.

[Academic]"Funeral Oration", Pericles (~400 B.C.)

[Academic]Chapters 19, 21, 22, 23, 25, *The Prince*, Machiavelli (1532)

10. STRATEGY

DATE: April 5
DUE: Reading. Assignment 3 is due.
TO DO: Start planning your next group. The final event.
WEEKLY QUESTION:
AGENDA:

READING:

[Academic][Chapter 2, 10] *The Innovator's Dilemma*, Clayton Christensen (Harvard Business Review Press, 2013).

[Academic][Chapter 1, 5] *Blue Ocean Strategy*, W. Chan Kim and Renee Marbourgh (Harvard Business Review Press, 2005).

[Academic]*Art of War*, Sun Tzu (1910 translation)

[Academic]"Book III: Of Strategy", *On War*, Carl von Clausewitz. (1832)

11. THE PSYCHOLOGY OF THE MARKETPLACE AND HOW TO ENTER IT

DATE: April 12
DUE: Reading.
TO DO: Final assignment coming soon.
WEEKLY QUESTION:
AGENDA:

READING:

[Academic][Chapter selections TBD] Inside the Tornado, Geoffrey Moore (Harper Business, 2005). The classic, recaps Crossing the Chasm then charts the next phases.

[Field] "A dozen things I've learned about marketing, distribution and sales", Tren Griffin (2014, *25iq)*

[Field] [Chapter selections TBD] Rules for Radicals, Saul Alinsky (Vintage, 1971). Leader of the 1950s community organizing movement that aided civil rights and civic reform in Chicago, NY, etc.

[Field] "The Why, How, and When of Sales", Mark Cranney of Andreessen Horowitz (audio)

[Field] "Coin-Operated Idiots", Lars Dalgaard of Andreessen Horowitz

12. MOVEMENTS AND SOCIAL CHANGE

DATE: April 19
DUE: Reading.
TO DO: Final assignment at last class.
WEEKLY QUESTION:
AGENDA:

READING:

[Field][Chapter selections pending] *The Autobiography of Margaret Sanger*, Margaret Sanger. (Dover, 2004). The founder of Planned Parenthood and advocate for women's rights tells her story.

[Academic][Chapter 1] Why David Sometimes Wins, Marshall Ganz (Oxford UP, 2010)

[Field] Nonviolent Struggle: 50 Crucial Points, Srdja Popovic et al. (Center for Applied Nonviolent Action and Strategies, 2007)

[Academic]"Letter from Birmingham Jail", Martin Luther King, Jr. (1963)

13. FINANCE AND MODELS AND FUNDRAISING

DATE: April 26
DUE: Reading.
TO DO: Final submission next week.
WEEKLY QUESTION:
AGENDA:

READING:

[Field][Chapters 3-7], *Venture Deals*, Brad Feld and Jason Mendelson (2012, Wiley). Legendary venture investor tells founders how to handle deals.

[Field][Chapters 2 and 7 of the original book], *High Tech Startup*, John Nesheim (2000, Free Press), the first data-driven detailed chronicler of patterns in financing and success of Silicon Valley across 1000s of companies and 100s of VCs.

[Field] "Fundraising Survival Guide", Paul Graham the founder of Ycombinator (2008)

14. THE COMPANY PITCH AND MAKING THE ARGUMENT FOR YOUR COMPANY

DATE: May 3
DUE: Final assignment.
TO DO: Good bye.
WEEKLY QUESTION:
AGENDA: We'll workshop company stories in person. Speak, defend, critique, and so on.

[Field] [Academic] Readings and videos: Steve Jobs iPhone launch, Barack Obama DNC 2004, Winston Churchill, June 4, 1940, House of Commons *("We Shall Fight on the Beaches")*, Teddy Roosevelt "The Man In the Arena", US Military "The long gray line", Elon Musk's Vision for the Future (with Steve Jurvetson, VC, at Stanford eCorner).

Columbia College Honor Code on Academic Integrity

https://www.college.columbia.edu/academics/academicintegrity

The Columbia College Student Council, on behalf of the whole student body, has resolved that maintaining academic integrity is the preserve of all members of our intellectual community – including and especially students.

As a consequence, all Columbia College students will now make the following pledge:

We, the undergraduate students of Columbia University, hereby pledge to value the integrity of our ideas and the ideas of others by honestly presenting our work, respecting authorship, and striving not simply for answers but for understanding in the pursuit of our common scholastic goals. In this way, we seek to build an academic community governed by our collective efforts, diligence, and Code of Honor.

In addition, all Columbia College students are committed to the following honor code:

I affirm that I will not plagiarize, use unauthorized materials, or give or receive illegitimate help on assignments, papers, or examinations. I will also uphold equity and honesty in the evaluation of my work and the work of others. I do so to sustain a community built around this Code of Honor.

Even More Content For Further Exploration

1. OPTIONAL:

"A Note on Vico's Concept of Knowledge", Isaiah Berlin (1969, NYRB). Essay considering the merit of Vico's critique of Descartes, that some knowledge is not deductive, formal, a priori and can never be.

"A Pragmatic Theory of Truth", William James in Reading for Philosophical Inquiry. by L and J Archie (2004, open source).

EXPECTED BACKGROUND:

Zero to One, Peter Thiel (Maybe?)

2. OPTIONAL:

The Art of the Really Long View, Peter Schwartz (Currency Doubleday, 1996). Futurist and scenario planning inventor describes the approach to high-uncertainty.
"Interview with Peter Schwartz", Long Now Foundation (podcast)

EXPECTED BACKGROUND:

Singularity is Near, Ray Kurzweil (Penguin, 2005). Inventor, futurist and Google advisor's sweeping prediction set and methodology for 2042.

3. OPTIONAL:

Practitioners and observers survey the progress
"This is when robots will start beating humans at every task" by Chris Weller (World Economic Forum, 2017)

"AI: Scary for the Right Reasons" by Vinod Khosla
"AI, Deep Learning, and Machine Learning: A Primer" by Frank Chen

Primary academic sources on milestones of progress
"Deep learning". LeCun, Bengio, Hinton (2015). Nature, 521(7553), pp. 436–444.
Neural Networks and Deep Learning, Michael Nielsen (Determination Press, 2015)

"A Proposal for the Dartmouth Summer Research Project on Artificial Intelligence", McCarthy, Minsky, Rochester, Shannon (*AI Magazine*, 1955).
"Computing machinery and intelligence". Alan Turing (1950, *Mind*, 59(236), pp. 433–460.) Foundational paper on AI.

Consequences
"The Commission on Work, Workers, and Technology" by Shift. Foundation-backed report on future of work.

More:
"AI Playbook" by Andreessen Horowitz
"Books, Papers, and Research" by Andreessen Horowitz
Deep Learning, Goodfellow, Bengio and Courville (2016, MIT Press)

4. OPTIONAL:

"Biology's Brave New World", Laurie Garrett (*Foreign Affairs*, 2013)

"Making the New Revolutions in Biology Safe", Laurie Garrett, (memo for Council on Foreign Relations, 2013)

"Biologist explains CRISPR at five levels of difficulty." Wired, 2015. (video)

"Sustainability: a meaty issue", Olive Heffernan, *Nature, 2017.*

"CRISPR: gene editing is just the beginning", Heidi Ledford, *Nature*, 2016.

5. OPTIONAL:

"3 ways good design makes you happy" by Don Norman (TED, video)

"Intro to the Design of Everyday Things" by Don Norman, online courseware, Udemy.

Inside Steve's Brain, Leander Kahney. Journalistic account of 1990s Apple.

"Coffee Break EP33 [VIDEO]: Design Thinking with Tim Brown" by Brenna Van Norman (video)

EXPECTED BACKGROUND:

Four Steps to the Epiphany, Steve Blank. The handbook for lean startup movement.

6. OPTIONAL:

The leaders visions of their products
Jeff Bezos, founder of Amazon.com, on the company vision in 1996: "amazon.com"
The iPhone launch presentation: "Steve Jobs introduces iPhone in 2007" by jschroter, 2007. (video)
LinkedIn's vision, circa 2015, from CEO Jeff Weiner: "LinkedIn Economic Graph | Economic Opportunity" by LinkedIn (video)

Perspectives on what the job of product management requires

"10x Not 10%" by Ken Norton of Google, on inventing 10x products.
"10x Not 10 % Product management by orders of magnitude" by Ken Norton (video)
"What Makes a Great Product Manger" by Lawrence Ripsher. Re: Horowitz by Pinterest PM.

"Find, Vet and Close the Best Product Managers" by Tod Jackson, PM from Dropbox.
"Hiring your first product manager" by Steven Sinofsky.
"What distinguishes the top 1% of product managers?" Quora comment by Amazon & Airbnb PM.
Jeff Weiner, CEO of LinkedIn on tools of the superstar PM on *LinkedIn Corporate Blog*
"The Product Manager's Lament", Eric Ries, 2008, *Startup Lessons Learned*.

Doing the job of product manager
"Kicking off product management workstreams", Ken Norton, 2008.
"Getting the right stuff done" Steven Sinosfky, 2015.
"Dealing with doubt and over-confidence in building something new", Steven Sinofsky, 2013.

How to know if you have it right
"A dozen lessons about Minimum Viable Product", Tren Griffin (2017, *25iq)*

Design thinking and product thinking in the wild
"Here's Why Juicero's Press is So Expensive", Ben Einstein (2017, *Bolt Medium*)
"CES 2016 Observations for Product People", Steven Sinofsky (2016, *Learning by Shipping*)
"My tablet has stickers", Steven Sinofsky, (2016, *Learning by Shipping*)
"iPad vs. Mac", Jean-Louis Gassee, 2017, *Monday Note.* Commentary on recent product evolution from Apple's original software engineering head.
"Memo to camera makers: Put Android in your device", Frederick Filloux, 2017, *Monday Note.*
"Apple Culture after 10 years of iPhone", J-L Gassee, 2017, *Monday Note.*

7. OPTIONAL:

"The Struggle", *The Hard Thing about Hard Things*, Ben Horowitz (2014, HarperBusiness). founder and leading venture capitalist.

How to Win Friends and Influence People, Dale Carnegie (Simon & Schuster, 1936).

"Alternative History of PayPal by Elon Musk", Elon Musk in *Gawker*. The mishmash and many roles of early stage.

My Inventions, Nikola Tesla (1919).

"The Rise and Fall of John DeLorean", Suzanne Snider (2006, *Longreads)*

8. OPTIONAL:

[To select TBD] "Selected Writings of Field Marshall Count Helmuth von Moltke on the Art of War", von Moltke (1891)

"Building a company where the best ideas win", Ray Dalio (2017, TED video).

"*Field Manual 6-22 Leader Development*", U.S. Army (2014).

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10. OPTIONAL:

The Asian Mind Game, Chin-ning Chu (Scribner, 1991). Guide to Sun-Tzu's art of war and Asian martial arts philosophy applied to business culture.

"You have discovered product market fit. What about a moat?", Tren Griffin (2017, *25iq*)

"Disruption's long slow complex journey", Steve Sinofsky (2016, *Learning by Shipping)*

11. OPTIONAL:

Influence, Robert Cialdini (Harper Business, 1984)

Basic nuts and bolts about marketing and new users, browse: Startup Marketing, Baremetrics, Guide to startup marketing, How to get your first ten customers.

12.

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13. OPTIONAL:

Venture Hacks collection, Naval Ravikant and Babak Nivi, cofounders of Angel List and serial investors and CEOs answer numerous pragmatic questions: What should I send investors? Part 1: Elevator Pitch, What should I send investors? Part 2: Deck, What should I send investors? Part 3: Business Plans, NDAs, and Traction, Grockit's founder on pitching investors, Is it safe to send my deck to investors?, High concept pitches for startups, From incoherent to high concept pitch, High concept pitch: Feedburner for iPhones, A Hollywood for startups, Elevator pitches with bullets, How Steve Jobs pitches a startup, How should I format my deck?, Why startup pitches fail (and how to fix them), Raising money is a black swan, Launch: Pitching Hacks, The Book, Our first online workshop: How to pitch investors, Pitching Hacks, in paperback, How to raise money without lying to investors, When you're raising money, the competition isn't the competition, "He has no clue whether it will be another Google, yet he has to make promises that only hucksters can make.", What's the vision?, Where's the demo?, Presentation Hacks, How much traction do I need?, If this is your first time raising money…, A conflict of interests

"Cap Tables, Share Structures, Valuations…" primer, Jason Rowley, (Crunchbase, 2017)

"The mechanics of venture capital", Marketplace radio, 2017. (audio)

"A dozen attributes of a scalable business", Tren Griffin (2017, *25iq)*

Even More Content That Didn't Make It
—————————————————————-

Other classical authors to consider
- Herodotus on Peloppenseian,
- Napoleon when he burns the ships at the Battle of the Nile,
- Sutonius The Lives of Caesars,
- Marcus Aurelius's Meditations
- Herodotus' Histories [Book 8]
- Xenophon's Anabasis
- Thucydides' History (lots of good stuff on war spread out throughout the whole text)
- Polybius' Histories (there's a lot on war there)
- Livy's Ab urbe condita (especially Book 30: battle of Zama)
- Vegetius' Epitoma rei militaris (probably the best extant classical work purely on military strategy)

Jeremy Rifkin, End of Work (Tarcher, 1996) link

Jeremy Rifkin, Zero Marginal Cost Society (St Martin's Griffin, 2015) link

Kevin Kelley, Inevitable

Industries of the Future, Alec Ross

interview with venture capitalist Steve Jurvetson (2016). (video at Stanford eCorner)
-
"Interview with Peter Thiel" (2005, Charlie Rose). Tries to predict the future in 2005. (video)

"Pursue Passions with a Vengeance", Melinda Gates at Stanford eCorner. (video)

Chaos Monkeys, Antonia Martinez (Harper, 2016)

First day to immediately blur the lines between social and commercial entrepreneurship.
"Millennial entrepreneurs driven by purpose as much as profits" by Deborah Nason (CNBC)

Doing Product Management day-to-day
https://www.kennorton.com/essays/leading-cross-functional-teams.html

The role of meetings
https://www.kennorton.com/essays/meetings-that-dont-suck.html

Death by Meetings
https://www.tablegroup.com/books/dbm

Extreme Ownership: Leadership Secrets of the Navy SEALs

Influence, Robert Cialdini
Poor Charlie's Almanac, psychology of misjudgment

Management - Tren Griffin
https://25iq.com/2015/07/05/a-dozen-things-ive-learned-from-ben-horowitz-about-management-investing-and-business/

Management - Sinofsky posts
https://blog.learningbyshipping.com/2013/07/30/lessons-from-org-structures/

Sports?

Radical Candor, pony mode and beast mode - exercise?

Failing cultures
- Failure stories - dysfunctional corporate environment book e.g., Business Adventures? Uber tell-all
- startup dysfunction stories
- Fowler on Uber
- Winning cultures (?)
Startup failure stories. The genre.
https://www.bloomberg.com/news/features/2017-09-08/inside-juicero-s-demise-from-prized-startup-to-fire-sale?curator=MediaREDEF
Eg stories written by the founders also

Disrupting and competing with the big guys - Sinofsky posts
https://blog.learningbyshipping.com/2016/01/29/5-ways-to-compete-with-big-incumbents/
https://blog.learningbyshipping.com/2013/10/03/disruption-and-woulda-coulda-shoulda/
https://blog.learningbyshipping.com/2015/05/23/the-lesson-of-dont-forget-all-the-parts-move/

https://25iq.com/2017/08/25/a-dozen-attributes-of-a-scalable-business/

Platform vs Powered By vs Product vs Feature
https://www.quora.com/What-is-the-difference-between-a-product-and-a-platform
https://hbr.org/2016/04/products-to-platforms-making-the-leap
https://www.forbes.com/sites/adrianbridgwater/2015/03/17/whats-the-difference-between-a-software-product-and-a-platform/#3754edb756a6